

Linear Regression Questions

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The following questions are based on the material discussed in the lecture slides on simple and multiple linear regression.

Question 1. The three goals of regression

A regression model may be built for prediction, explanation of association, or causal inference.

- (a) Explain the difference between these three goals.
- (b) Give one example where the main goal is prediction.
- (c) Give one example where the main goal is explanation of association.
- (d) Why should causal language be used carefully in observational data?

Question 2. What does a linear regression model predict?

Consider the simple linear regression model

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad \mathbb{E}(\varepsilon \mid X) = 0.$$

- (a) What does the regression line represent?
- (b) What is meant by the conditional mean $\mathbb{E}(Y \mid X = x)$?
- (c) Why do several observations with the same predictor value x not necessarily have the same response Y ?
- (d) Define the residual for the i th observation.

Question 3. Simple linear regression output: apartment rent and apartment size

A study examines whether monthly rent depends on apartment size. The response variable is monthly rent (in euros), and the predictor is apartment size (in square meters).

A simple linear regression gives the following output:

	Coefficient	Std. Error	t -statistic	p -value
Intercept	220.5	38.4	5.74	< 0.001
Size	9.8	1.1	8.91	< 0.001

The 95% confidence interval for the slope is

$$[7.6, 12.0].$$

- (a) Interpret the slope coefficient carefully, including units.
- (b) Interpret the intercept. Why may this interpretation be questionable?

- (c) State the null and alternative hypotheses for testing whether apartment size is associated with rent.
- (d) What conclusion would you make from the p -value?
- (e) What does the confidence interval tell us about the slope?

Question 4. Residual sum of squares and least squares

- (a) What is the residual sum of squares (RSS) in simple linear regression?
- (b) Why does the least-squares method choose $\hat{\beta}_0$ and $\hat{\beta}_1$ by minimizing RSS?
- (c) Explain in words what it means for a fitted line to have a smaller RSS than another fitted line.

Question 5. Confidence interval versus prediction interval

A company fits a simple linear regression model to predict the daily number of customers from the amount spent on local advertising.

For a certain advertising budget, the fitted model gives: $\hat{y} = 84$.

A 95% confidence interval for the mean response is $[79, 89]$, and a 95% prediction interval for one new day is $[60, 108]$.

- (a) Explain the difference between the confidence interval and the prediction interval.
- (b) Which interval is wider, and why?
- (c) Which interval is relevant if we want to estimate the average number of customers for all days with this advertising budget?
- (d) Which interval is relevant if we want to predict the number of customers on one particular future day?
- (e) Why are predictions outside the observed range of advertising budgets less reliable?

Question 6. Assessing the overall accuracy of a simple regression model

For the same model relating apartment rent to apartment size, suppose we also have:

Quantity	Value
Residual Standard Error	82.4
R^2	0.58
F -statistic	79.4

- (a) Interpret the residual standard error in words.
- (b) Interpret $R^2 = 0.58$.
- (c) Why should R^2 not be used alone to judge the quality of the model?
- (d) What does a large F -statistic suggest in this setting?
- (e) Suppose the correlation between rent and size is $r = 0.76$. How is this related to R^2 in simple linear regression?

Question 7. Multiple linear regression and interpretation “holding others fixed”

A model is fitted to predict the final course score of a student using: Homework points, Attendance percentage, Hours of independent study per week.

The fitted model is:

$$\widehat{\text{Score}} = 18.2 + 0.45(\text{Homework}) + 0.22(\text{Attendance}) + 1.6(\text{StudyHours}).$$

- (a) Interpret the coefficient of Homework points.
- (b) Interpret the coefficient of Attendance percentage.
- (c) Interpret the coefficient of StudyHours.
- (d) In each interpretation above, what does “holding the other predictors fixed” mean?
- (e) Why can interpretation become difficult when predictors are strongly correlated?

Question 8. Multiple regression output: hospital recovery time

A study models recovery time after a minor procedure (in days) using patient age, number of exercise sessions per week, and average hours of sleep per night.

	Coefficient	Std. Error	<i>t</i> -statistic	<i>p</i> -value
Intercept	12.4	1.8	6.89	< 0.001
Age	0.09	0.02	4.50	< 0.001
Exercise	-0.70	0.21	-3.33	0.0014
Sleep	-0.15	0.24	-0.63	0.5310

- (a) Interpret the coefficient of Age.
- (b) Interpret the coefficient of Exercise.
- (c) What does the coefficient of Sleep suggest in this fitted model?
- (d) Does the large *p*-value for Sleep prove that sleep is completely unrelated to recovery? Explain carefully.
- (e) State the null and alternative hypotheses for testing whether the Exercise coefficient is zero.
- (f) Should we add additional predictors in models? How do we evaluate which model is better?

Question 9. Qualitative predictors and dummy variables

Suppose we model monthly phone bill Y using monthly data usage and subscription type. The qualitative predictor Premium is coded as

$$\text{Premium}_i = \begin{cases} 0, & \text{if standard plan,} \\ 1, & \text{if premium plan.} \end{cases}$$

The fitted model is:

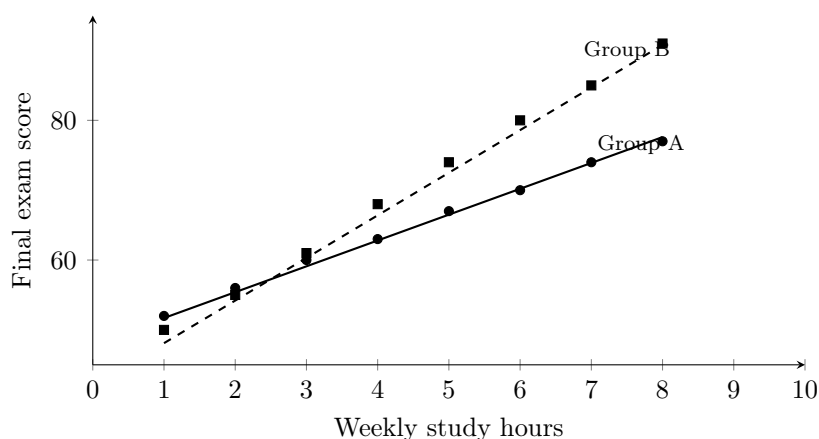
$$Y_i \approx \beta_0 + \beta_1 \text{Usage}_i + \beta_2 \text{Premium}_i.$$

- (a) Write the fitted regression equation separately for standard-plan users and premium-plan users.

- (b) What does β_2 represent in this model?
- (c) What does it mean that the two regression lines are parallel? How do we know this is a suitable model?
- (d) Which group is the baseline group?
- (e) If $\beta_2 > 0$, how do you interpret that?
- (f) When is it suitable to use dummy variables for modeling?

Question 10. Graphical interpretation and interaction

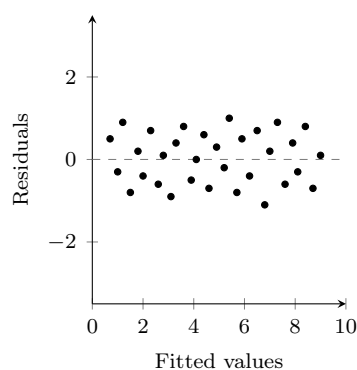
Consider the following figure showing the relationship between weekly study hours and final exam score for two groups of students.



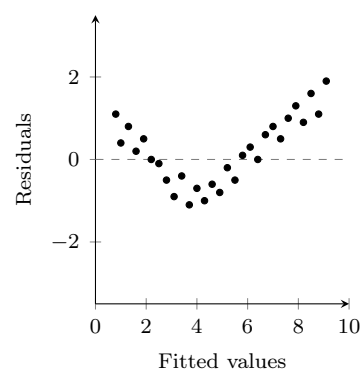
- (a) Do the two groups appear to have the same slope?
- (b) Would a model with only a dummy variable for group? Explain.
- (c) What does this graph suggest about the need for an interaction term?
- (d) What would a positive interaction coefficient mean in this setting?

Question 11. Residual plots: what do they show, and what should we do next?

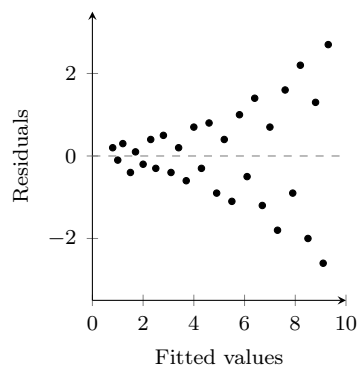
A data analyst fits three different linear regression models. For each model, a residual-versus-fitted plot is shown below.



Plot A



Plot B



Plot C

- For each plot, describe what you observe.
- Which plot looks most consistent with a reasonable linear regression fit?
- Which plot suggests that the relationship may not be adequately modeled by a straight line?
- Which plot suggests non-constant variance of the errors?
- For Plot B, what would be a sensible next step in the modeling?
- For Plot C, what practical caution or next step would you suggest?

Question 12. A significant coefficient, but a small R^2

A health researcher studies whether daily walking time (in minutes) is associated with systolic blood pressure. A simple linear regression model is fitted using data from 300 adults.

	Coefficient	Std. Error	t -statistic	p -value
Intercept	142.8	2.9	49.24	< 0.001
Walking time	-0.18	0.05	-3.60	< 0.001

Additional information:

$$R^2 = 0.042.$$

- Interpret the slope coefficient, including units.
- State the null and alternative hypotheses for testing whether walking time is associated with systolic blood pressure.
- What conclusion would you make from the reported p -value?
- Interpret $R^2 = 0.042$.
- Explain how it is possible for the slope to be statistically significant even though R^2 is small.
- Does this result suggest a strong predictive relationship? Explain.

Question 13. The overall F -test in multiple regression

In multiple linear regression, we may first ask whether the regression as a whole explains anything at all.

- State the null and alternative hypotheses for the overall F -test in a multiple regression model with predictors $X_1, \dots, X_p$.
- What does it mean if the p -value of the F -test is very small?

- (c) If the overall F -test is significant, what can we conclude?
- (d) If the overall F -test is significant, what can we *not* conclude yet?
- (e) After finding that the overall F -test is significant, what is the next natural question in the analysis?

Question 14. Why can the conclusion change from simple to multiple regression?

Suppose that in a simple linear regression, predictor X_3 has a significant positive coefficient. However, after fitting a multiple regression with predictors X_1 , X_2 , and X_3 , the coefficient of X_3 is no longer significant.

- (a) Give one possible explanation for why this can happen.
- (b) What does it mean to interpret the coefficient of X_3 in the multiple regression *holding the other predictors fixed*?
- (c) How can correlation among predictors affect this interpretation?
- (d) Does the loss of significance in the multiple regression prove that X_3 has no causal effect? Explain carefully.
- (e) Why is this kind of comparison between simple and multiple regression important in practice?

Question 15. Simpson's paradox and subgroup structure

A trend in aggregated data may change or even reverse after the data are split into meaningful groups.

- (a) What is Simpson's paradox?
- (b) Why can an aggregate trend be misleading?
- (c) What role does a third variable play in Simpson's paradox?
- (d) Why is Simpson's paradox important in regression analysis?
- (e) What should an analyst do if they suspect that the overall association may be hiding different subgroup patterns?
- (f) Why is it dangerous to interpret only the aggregate regression line without checking subgroup structure?